BIO204 2nd Mid semester Exam (2026) IISER Mohali

Date: 07/03/2026 **Duration:** 60 min

Maximum marks: 25

Instructions:

a. All questions are compulsory. Each question in Sections A, B, C carries 1 mark.

Sections A

1. You are setting up a PCR reaction. Accidentally, instead of dNTPs, you added ddNTPs to the mixture. Now, in the results, you would observe:
   a. Non-specific amplification
   b. Specific amplification of the desired size
   c. Products of different lengths
   d. No amplification
2. A researcher was cloning a gene of interest into a vector. She transformed the ligated DNA products into *E. coli*. However, two types of transformants were obtained: vector + insert and vector self-ligated. Which one of the following methods cannot be used to identify the correctly cloned construct?
   a. PCR
   b. Restriction digestion
   c. DNA sequencing
   d. DNA ligation
3. What is the origin of meganucleases in DNA?
   a. Promoter elements
   b. Terminator elements
   c. Intronic regions
   d. Exonic regions
4. Which one among the following DNA-cutting tools is easiest to engineer?
   a. Zinc Finger Nucleases
   b. Meganucleases
   c. Transcription Activator-Like Effector Nucleases
5. In Zinc Finger Nucleases, five zinc fingers are designed for each of the left and right zinc finger proteins. What would be the nucleotide sequence specificity of such a zinc finger nuclease?
   a. 25 nucleotides
   b. 30 nucleotides
   c. 5 nucleotides
   d. 10 nucleotides
6. In Ramachandran plot, amino acid showing most allowed conformations is:
   a. Alanine
   b. Glycine
   c. Serine
   d. Proline
7. Which program is used for secondary structure assignment in proteins:
   a. DSSP
   b. PSIPRED
   c. BLAST
   d. CLUSTALW
8. Secondary structure in protein is stabilized by:
   a. Disulfide bond
   b. Hydrogen bond
   c. Electrostatic bond
   d. Phi-psi angles
9. Which amino acid can form stable left-handed alpha-helix structure:
   a. Alanine
   b. Serine
   c. Proline
   d. Glycine
10. A researcher found 0° omega (ω) torsion angle of an unknown residue in a peptide bond. This unknown residue is likely:
    a. Glycine
    b. Tryptophan
    c. Proline
    d. Tyrosine

Section B (Answer in One Line / Fill in the blanks)

11. A researcher observed that during PCR amplification, several non-specific products are amplified in addition to the desired product. He has asked you to help him. According to you, what could be the reason? (One-line answer for both options)
    a. ***This is because primers annealed to non-desired sites as well which resulted in non-specific bands amplification in addition to the desired product.***
    b. ***He should change, increase the primer lengths, and/or increase the annealing temperature.***
12. Why do we not use *E. coli* DNA polymerase in PCR reactions? ***We do not use E. coli DNA polymerase because it is not thermostable and will be denatured under the high-temperature conditions of PCR.***
13. If we use DNA polymerase from *Thermus aquaticus* in PCR, why do we not use DNA ligase from *Thermus aquaticus* for routine DNA product ligation? ***We do not use Thermus aquaticus DNA ligase, as the enzyme's optimal activity occurs at higher temperatures and therefore it does not ligate efficiently in routine ligation reactions.***
14. If a restriction enzyme has a sequence specificity of 8 nucleotides, then in 100 kb of randomly distributed DNA sequence, how many sites are theoretically expected? ***~1.5 sites (approximately 1–2 sites).***
15. What does "RVDs" in Transcription Activator-Like Effector Nucleases stand for? ***Repeat Variable Diresidues***
16. Name the Cas proteins that help in the protospacer adaptation process in a CRISPR-Cas system. ***Cas1 and Cas2***
17. Since RNA sequence in CRISPR is complementary to the bacterial DNA from which it is derived, why is the genomic DNA not cleaved in the CRISPR-Cas system? ***Because bacteria genome lacks PAM sequences next to the spacer region.***
18. After a double-stranded DNA break, the NHEJ pathway generally leads to gene knockdown because of ***insertion and deletion***. (Two words only)
19. Why L-amino acids do not readily adopt a left-handed helical conformation? ***Due to steric clash between backbone CO/NH with the side -chain CBeta***
20. he phi and psi torsion angle in protein is defined by $C_{i-1}N_iC\alpha_iC_i$ and $N_iC\alpha_iC_iN_{i+1}$ respectively.
21. A researcher is validating two models of the same protein. Ramachandran plot statistics are for models A and B are as follows:
    Model A: 96% of residues lie in allowed regions, and 4% in disallowed regions.
    Model B: 99% of residues are in allowed regions, and 1% in disallowed region.
    Based on these plots, which model is generally considered a better quality structure? Justify.
    Model B as it has more residues lying in the allowed region leading to least steric clashes.

Section C (State True/False)

22. In engineered genome editing tools such as Zinc Finger Nucleases and Transcription Activator-Like Effector Nucleases, the FokI cleavage domain is used because it is highly sequence-specific in its cleavage activity. ***False*** (True or False)

23. Base pairing of primers with the template is essential at the 5′ end of the primer, but not at the 3′ end. ***False*** (True or False?)
24. If we plot Ramachandran map separately for each amino acids (except G and P residues) all will exhibit similar allowed and disallowed regions. ***True*** (True/False)
25. Among helical structures in proteins, 3_{10}-helix is the most abundant secondary structure ***False*** (True/False)